

Study Disaster Management Datasets to Identify Regions with High Disaster Vulnerability

Name: Mohammed Mudhassir A

Reg No: 192424367

Assignment No. 5

1. Problem Understanding

Disaster management authorities need to identify regions that are highly vulnerable to disasters such as floods, cyclones, earthquakes, landslides, and droughts. Historical disaster datasets can be analyzed using **NumPy and Pandas** to identify high-risk regions.

The objective of this analysis is to:

- Clean and preprocess disaster data.
 - Calculate disaster-related statistics.
 - Group data by region.
 - Identify regions with high disaster frequency and impact.
 - Rank regions according to a vulnerability score.
 - Visualize the results.
 - Provide useful insights for disaster preparedness.
-

2. Dataset

A dataset can contain the following attributes:

Column	Description
Region	Name of the region
Disaster_Type	Flood, Cyclone, Earthquake, etc.
Year	Year of occurrence

Column	Description
Disaster_Count	Number of disasters
Population_Affected	Number of affected people
Economic_Loss	Financial loss
Rainfall	Rainfall during the period
Temperature	Average temperature
Infrastructure_Index	Infrastructure quality
Vulnerability_Score	Calculated vulnerability

Sample Data

Region	Disaster Type	Year	Disaster Count	Population Affected	Economic Loss	Infrastructure
Chennai	Flood	2024	5	85000	120	55
Mumbai	Flood	2024	4	72000	150	65
Delhi	Flood	2024	3	45000	90	70
Kolkata	Cyclone	2024	6	110000	180	50
Hyderabad	Flood	2024	2	25000	60	75

3. Data Loading

Use Pandas to read the dataset.

```
import pandas as pd
```

```
import numpy as np
```

```
df = pd.read_csv("disaster_management.csv")
```

```
print(df.head())
```

```
print(df.info())
```

Check missing values:

```
print(df.isnull().sum())
```

4. Data Preprocessing

Remove duplicate records:

```
df = df.drop_duplicates()
```

Handle missing numerical values:

```
numeric_columns = [  
    "Disaster_Count",  
    "Population_Affected",  
    "Economic_Loss",  
    "Rainfall",  
    "Temperature",  
    "Infrastructure_Index"  
]
```

for col in numeric_columns:

```
    df[col] = df[col].fillna(df[col].median())
```

Standardize region names:

```
df["Region"] = (  
    df["Region"]
```

```
.str.strip()

.str.title()

)
```

Check invalid values:

```
df = df[df["Disaster_Count"] >= 0]

df = df[df["Population_Affected"] >= 0]

df = df[df["Economic_Loss"] >= 0]
```

5. Statistical Analysis

Calculate basic statistics:

```
print(df[numeric_columns].describe())
```

Calculate the average number of disasters:

```
average_disasters = df["Disaster_Count"].mean()

print("Average disasters:", average_disasters)
```

Maximum number of disasters:

```
maximum = df["Disaster_Count"].max()

print("Maximum disasters:", maximum)
```

Total affected population:

```
total_affected = df["Population_Affected"].sum()

print("Total affected:", total_affected)
```

6. Grouping by Region

Group the dataset by region:

```
region_summary = df.groupby("Region").agg(
```

```

Total_Disasters=("Disaster_Count", "sum"),
Average_Disasters=("Disaster_Count", "mean"),
Total_Affected=("Population_Affected", "sum"),
Total_Loss=("Economic_Loss", "sum")
)

```

```
print(region_summary)
```

This provides a regional summary of disaster frequency and impact.

7. Calculating Vulnerability Score

A simple vulnerability score can be created using normalized factors.

For example:

where:

- = normalized disaster frequency
- = normalized affected population
- = normalized economic loss
- = infrastructure risk

First normalize the values:

```

region_summary["Disaster_Score"] = (
    region_summary["Total_Disasters"] /
    region_summary["Total_Disasters"].max()
)

```

```
region_summary["Population_Score"] = (
```

```
region_summary["Total_Affected"] /  
region_summary["Total_Affected"].max()  
)
```

```
region_summary["Loss_Score"] = (  
    region_summary["Total_Loss"] /  
    region_summary["Total_Loss"].max()  
)
```

Calculate the final score:

```
region_summary["Vulnerability_Score"] = (  
    0.4 * region_summary["Disaster_Score"] +  
    0.3 * region_summary["Population_Score"] +  
    0.3 * region_summary["Loss_Score"]  
)
```

8. Identify High-Vulnerability Regions

Sort regions according to vulnerability:

```
ranked_regions = region_summary.sort_values(  
    "Vulnerability_Score",  
    ascending=False  
)
```

```
print(ranked_regions)
```

Select high-risk regions:

```
high_risk = ranked_regions[
    ranked_regions["Vulnerability_Score"] >= 0.7
]
```

```
print("High Vulnerability Regions:")
print(high_risk)
```

Thus, the regions with the highest scores can be prioritized for disaster-management activities.

9. Disaster-Type Analysis

Group the data by disaster type:

```
disaster_summary = df.groupby(
    "Disaster_Type"
)["Disaster_Count"].sum()
```

```
print(disaster_summary)
```

This identifies which type of disaster occurs most frequently.

For example:

Flood → 35

Cyclone → 22

Earthquake → 15

Landslide → 10

This information can help authorities prepare for the most frequent disaster types.

10. Visualization

Bar Chart – Disaster Vulnerability by Region

```
import matplotlib.pyplot as plt

ranked_regions["Vulnerability_Score"].plot(

    kind="bar",

    figsize=(10, 5)

)

plt.title("Disaster Vulnerability by Region")

plt.xlabel("Region")

plt.ylabel("Vulnerability Score")

plt.xticks(rotation=45)

plt.tight_layout()

plt.show()
```

Disaster Frequency

```
disaster_summary.plot(

    kind="bar",

    figsize=(8, 5)

)

plt.title("Disaster Frequency by Type")

plt.xlabel("Disaster Type")

plt.ylabel("Total Disasters")

plt.tight_layout()
```



```
plt.show()
```

Economic Loss

```
region_summary["Total_Loss"].sort_values(  
    ascending=False  
).plot(kind="bar")
```

```
plt.title("Economic Loss by Region")
```

```
plt.xlabel("Region")
```

```
plt.ylabel("Economic Loss")
```

```
plt.show()
```

11. Interpretation of Results

After analyzing the dataset, the following types of conclusions can be obtained:

- Regions with **high disaster frequency** require stronger disaster-preparedness systems.
 - Regions with a **large affected population** need better evacuation and emergency-response facilities.
 - High economic losses indicate areas where disaster-resistant infrastructure should be prioritized.
 - Regions with low infrastructure quality may have greater vulnerability.
 - Flood- or cyclone-prone regions may require specialized early-warning systems.
 - The ranking can help governments prioritize limited disaster-management resources.
-

12. Innovation

A useful extension is a **Disaster Vulnerability Dashboard**.

The dashboard could display:

REGION: Chennai

Disaster Frequency: High

Population Risk: High

Economic Loss: Medium

Infrastructure: Medium

Vulnerability Score: 82%

Risk Level: HIGH

Recommended Actions:

- ✓ Improve early warning systems
- ✓ Strengthen drainage infrastructure
- ✓ Prepare emergency shelters
- ✓ Increase disaster-response resources

A map-based visualization could also show regions according to their vulnerability levels.

13. SDG Mapping

SDG 11 – Sustainable Cities and Communities

This analysis supports **SDG 11** by helping identify vulnerable communities and supporting:

- Disaster-resilient infrastructure.
- Safer urban planning.
- Emergency preparedness.
- Resource allocation.

- Protection of vulnerable populations.

SDG 13 – Climate Action

The analysis supports **SDG 13** by helping organizations:

- Understand climate-related disaster patterns.
 - Identify high-risk regions.
 - Improve climate adaptation.
 - Develop disaster-risk reduction strategies.
 - Support evidence-based climate planning.
-

14. Expected Outcome

The analysis produces a **ranked list of regions according to disaster vulnerability**.

Example:

Rank	Region	Disaster Score	Population Risk	Economic Loss	Vulnerability	Risk
1	Kolkata	0.95	0.92	0.88	0.92	High
2	Chennai	0.85	0.80	0.72	0.80	High
3	Mumbai	0.72	0.68	0.75	0.72	High
4	Delhi	0.55	0.48	0.50	0.52	Medium
5	Hyderabad	0.35	0.30	0.32	0.33	Low

The exact ranking will depend on the actual dataset.

15. Conclusion

Using **NumPy** and **Pandas**, historical disaster-management data can be cleaned, filtered, grouped, aggregated, and analyzed efficiently. A vulnerability score can then be calculated to

rank regions according to disaster frequency, population impact, and economic losses. Visualization makes the results easier to understand and supports decision-making.

The proposed system can help authorities **identify high-risk regions, prioritize disaster-management resources, improve community resilience, and support climate adaptation**, making it directly relevant to **SDG 11 and SDG 13**.